

EE-102 Electric Circuit Analysis

Experiment # 4

To Investigate Characteristics of DC Parallel Circuit

Objective/s:

- To Investigate Characteristics of DC Parallel Circuit

Apparatus

- Digital Multimeter (DMM),
- Power Supply,
- Wire Stripper,
- Resistors: 100Ω, 3.6kΩ, 4.7kΩ, 6.8kΩ, 10kΩ

1. Parallel Circuit

In a parallel circuit (Fig4.1) the voltage across parallel elements is the same.

The total or equivalent resistance (R_T) is given by.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}$$

If there are only two resistors in parallel, it is more convenient to use.

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

In any case, the total resistance will **always be less** than the resistance of the smallest resistor of the parallel network.

For the network of Fig 4.1

The currents are related by the following expression.

$$I_T = I_1 + I_2 + I_3 + \dots + I_N$$

Applying current divider rule (CDR) & the network of Fig 4.2

$$I_1 = \frac{I_T R_2}{R_1 + R_2} \quad \text{and} \quad I_2 = \frac{I_T R_1}{R_1 + R_2}$$

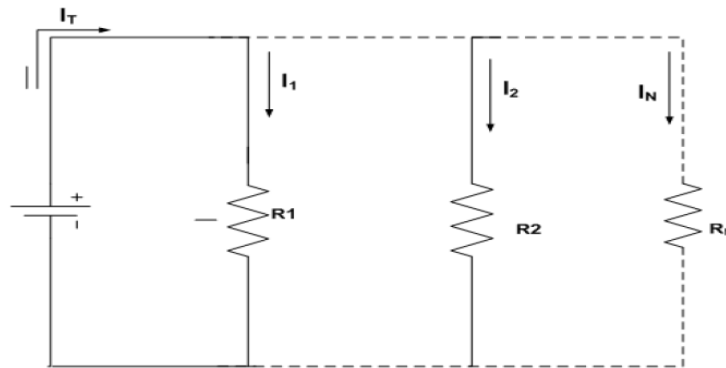


Figure 1

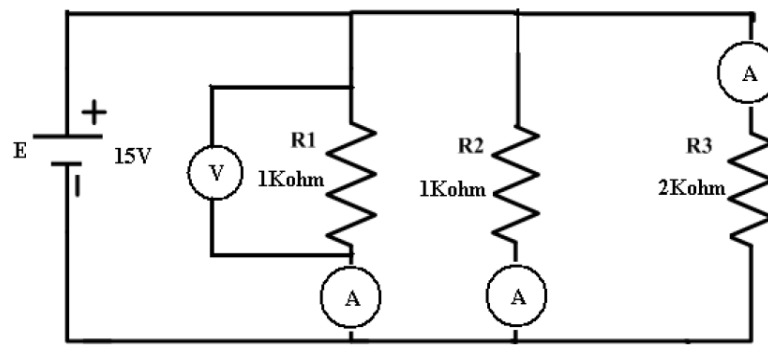


Figure 2

For equal parallel resistors, the current divides equally and the total resistance is the value of one divided by the 'N' number of equal parallel resistors, i.e

$$R_T = \frac{R}{N}$$

For a parallel combination of **N** resistors, the current **IK** through **RK** is.

$$I_K = I_T \times \frac{\frac{1}{R_K}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}}$$

2. Procedure

1. Construct the circuit shown in Fig 1.
2. Set the DC supply to 15V by using DMM. Pick the resistances values 1KΩ, 1KΩ, and 2KΩ. Also verify their resistance by using DMM.

3. Measure voltage across each resistor with DMM and record it in the Table b.
4. Measure the currents I_T , I_1 , I_2 , and I_3 .
5. Shut down & disconnect the power supply. Then measure input resistance ' R_T ' across points A-B using DMM. Record that value.
6. Now calculate respective voltages (using $V=IR$) and R_T
7. Calculate I_1 , I_2 , I_3 using CDR.
8. Create an open circuit by removing R_2 and measure all voltages and currents.

3. Tasks

1. Resistor

S.No.	Nominal Values (Ω)	Measured Value (Ω)	R_T (Measured) (Ω)	R_T (Measured) (Ω)
1.	$R_1 = 1k$			
2.	$R_2 = 1k$			
3.	$R_3 = 2k$			

Table 1

2. Voltage

S.No.		Measured Values (V)	Calculated Value (V) (Ohm's Law)	Measured Value when R_2 is Open Circuited(V)
1.	V_1			
2.	V_2			
3.	V_3			

Table 2

3. Current

S.No.	Measured Values (A)	Calculated Value (A) (CDR)	Measured Value (A) when R_2 is Open Circuited
1.	$I_1 =$	$I_1 =$	$I_1 =$
2.	$I_2 =$	$I_2 =$	$I_2 =$
3.	$I_3 =$	$I_3 =$	$I_3 =$
4.	$I_T =$		$I_T =$

Table 3